Discuss: Who is correct? Is Mohan really in motion?

We can clearly say that both Reka and Babu are correct. From the point of view of Babu, Mohan along with the bus is in motion; but for Reka who is sitting beside him, he is at one place; therefore stationary. So, according to Babu, Mohan is in motion; Mohan is at rest from Reka's observation. Can you think any other examples?



Answer by observing the situation in the picture

Event 1: The man in the boat is **moving** with respect to the bank of river. He is at **rest** with respect to the boat.



Event 2:

The girl on the swing is _____ with respect to the seat of the swing.

She is _____ with respect to the garden.



Event 3: Nisha is going to her grandmother's house by bicycle. Sitting on the bicycle, Nisha is _____ with respect to the road.



She is _____ with respect to the bicycle.

Take the case of a book on a table at rest. Is it really without any motion? We know that Earth is rotating on its axis; therefore the table along with the book must be rotating. Is it not? We are also moving along with the earth. Therefore, from the point of view of the ground on which we stand, the book is at 'rest'. Similarly, while travelling in a bus, we feel that the poles and trees seem to move backwards, and the things inside the bus are stationary.



An object may appear to be stationary for one observer and appear to be moving for another. An object is at rest in relation to a certain set of objects and moving in relation to another set of objects. **This implies that rest and motion are relative.**